

NADI 2025: The First Multidialectal Arabic Speech Processing Shared Task

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Abstract

We present the findings of the sixth Nuanced Arabic Dialect Identification (NADI 2025) Shared Task, which focused on Arabic speech dialect processing across three subtasks: spoken dialect identification (Subtask 1), speech recognition (Subtask 2), and diacritic restoration for spoken dialects (Subtask 3). A total of 44 teams registered, and during the testing phase, 100 valid submissions were received from eight unique teams. The distribution was as follows: 34 submissions for Subtask 1 “five teams”, 47 submissions for Subtask 2 “six teams”, and 19 submissions for Subtask 3 “two teams”. The best-performing systems achieved 79.8% accuracy on Subtask 1, 35.68/12.20 WER/CER (overall average) on Subtask 2, and 55/13 WER/CER on Subtask 3. These results highlight the ongoing challenges of Arabic dialect speech processing, particularly in dialect identification, recognition, and diacritic restoration. We also summarize the methods adopted by participating teams and briefly outline directions for future editions of NADI.¹

1 Introduction

Spoken Arabic exhibits a remarkable degree of linguistic diversity. Beyond Modern Standard Arabic (MSA) and Classical Arabic (CA), which have historically dominated computational work, Arabic encompasses numerous regional and national dialects that differ across all linguistic levels (phonology, morphology, lexicon, and syntax) and in discourse/pragmatics (Talafha et al., 2024; Jarrar et al., 2023). These varieties also frequently exhibit intra- and inter-sentential code-switching with other languages (Abdul-Mageed et al., 2024). These varieties dominate everyday communication across the Arab world yet remain under-represented in annotated datasets and resources (Bouamor et al.,

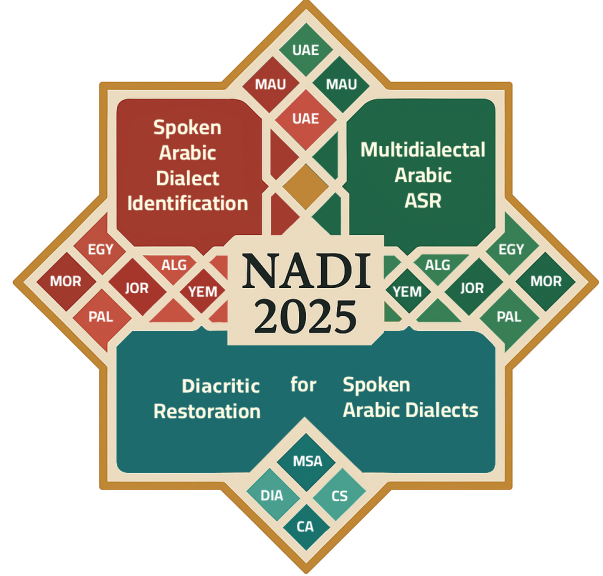


Figure 1: Overview of the NADI 2025 shared tasks.

2018; Darwish et al., 2021; Abdul-Mageed et al., 2020, 2023). At the same time, many downstream applications—from automatic transcription and virtual assistants to text-to-speech and educational tools—depend on accurate handling of dialectal speech and the diacritics that indicate short vowels and phonological features. Existing systems trained on CA/MSA (Elmadany et al., 2023a; Toyin et al., 2023) often ignore these diacritics or assume text forms, leaving a large gap between technology and real-world usage.

NADI shared task series, hosted at the Arabic-NLP conference² since 2020, was created to alleviate this bottleneck by providing curated datasets and standardized evaluation settings for dialect identification, translation and related tasks (Abdul-Mageed et al., 2020, 2021; Abdul-Mageed et al., 2022, 2023, 2024). These earlier, text-focused editions—together with the general observation that

¹The official leaderboards and datasets for NADI 2025 are available at <https://nadi.dlnlp.ai/2025>.

²Formerly the Workshop on Arabic Natural Language Processing, WANLP

Arabic dialects remain under-studied due to limited resources—motivate a shift in NADI 2025 toward speech and diacritization.

NADI 2025 marks the *sixth* edition of the NADI shared task series, hosted by the Third Arabic Natural Language Processing Conference (ArabicNLP 2025³). In the following, we introduce several key new features that set it apart from previous versions, focusing on the challenges of real-world, spoken Arabic dialects:

A unified speech processing benchmark. This edition brings together three distinct but complementary tasks, “*dialect identification*”, “*automatic speech recognition*”, and “*diacritic restoration*”, under one umbrella. This creates a comprehensive benchmark for evaluating system performance across the full spectrum of challenges in Arabic speech processing.

New evaluation datasets and unified benchmarking framework. We introduce a comprehensive suite of newly-curated datasets across all three subtasks. This includes a high-quality blind test sets *eight-hours* speech corpus for spoken dialect identification, a large-scale 10, 807 *utterances* for ASR, and a 1, 332 *utterances* for diacritic restoration, all covering diverse Arabic varieties. Beyond the data itself, NADI 2025 establishes a robust and unified evaluation framework featuring large-scale blind test sets to ensure fair comparison. This framework introduces novel paradigms, such as benchmarking model *adaptation* in the ADI task and offering distinct *open* and *closed* tracks for Diacritic Restoration.

A novel diacritic restoration task. We introduce the first shared task for diacritic restoration that moves beyond formal written Arabic (CA and MSA) to target *spoken dialects* and *code-switched language*. The task is uniquely designed to encourage multimodal solutions that leverage both speech and text as input.

Figure 1 provides a schematic overview of the NADI 2025 shared task, illustrating its three main subtasks including **Spoken Arabic Dialect Identification**, which covers *eight* regional dialects as “*Algerian*” (ALG), “*Egyptian*” (EGY), “*Emirati*” (UAE), “*Jordanian*” (JOR), “*Mauritanian*” (MAU), “*Moroccan*” (MOR), “*Palestinian*” (PAL), and “*Yemeni*” (YEM); **Multidialectal Arabic ASR**, which targets the exact same set of dialects; and **Diacritic Restoration for Spoken Arabic Dialects**,

which encompasses MSA, mixed dialects, code-switched varieties, and CA.

The rest of the paper is organized as follows: Section 2 provides a review of related work on spoken Arabic processing and the history of the NADI shared task. In Section 3, we describe the NADI 2025 shared task in detail, including the three subtasks, their datasets, and evaluation metrics. Section 4 presents the results for all participating teams and baselines, followed by an overview of the submitted systems in Section 5. We conclude the paper in Section 6.

2 Literature Review

Unlike previous NADI tasks that relied on text, NADI 2025 concentrates on spoken Arabic dialects. Accordingly, this section covers related work on the subtasks of spoken language identification, ASR, and diacritic restoration. Before delving into the related work, it is useful to explore the history of NADI and its growth since its inception.

2.1 NADI Shared Task: Origins and Growth

NADI-2020, the first NADI shared task (Abdul-Mageed et al., 2020) involved two subtasks, one targeting country level (21 countries) and another focusing on province level (100 provinces), both exploiting X, *formerly Twitter*, data. NADI 2020 was the first shared task to exploit naturally occurring fine-grained dialectal text at the sub-country level.

NADI-2021, the second version (Abdul-Mageed et al., 2021) targeted the same 21 Arab countries and 100 corresponding provinces as NADI 2020, also using X data. However, it improved upon the previous version by removing non-Arabic data and distinguishing between MSA and dialectal Arabic (DA). It involved four subtasks: MSA-country, DA-country, MSA-province, and DA-province.

NADI-2022 (Abdul-Mageed et al., 2022) continued the focus on studying Arabic dialects at the country level, but also included dialectal sentiment analysis with an objective to explore variation in socio-geographical regions that had not been extensively studied before.

NADI-2023, the fourth edition (Abdul-Mageed et al., 2023), proposed new machine translation subtasks from four dialectal Arabic varieties to MSA, in two themes (open-track and closed-track) as well as a dialect identification subtask at the country level.

³<https://arabicnlp2025.sigarab.org/>

Finally, NADI-2024, the fifth edition (Abdul-Mageed et al., 2024), targeted both dialect ID cast as a multi-label task, identification of the Arabic level of dialectness, and dialect-to-MSA machine translation.

2.2 Spoken Dialect Identification

Although CA and MSA have been extensively examined (Harrell, 1962; Badawi, 1973; Brustad, 2000; Holes, 2004), dialectal Arabic (DA) became the center of attention only relatively recently. A significant challenge in studying DA has been the scarcity of resources, prompting researchers to create new DA datasets targeting limited regions (Gadalla et al., 1997; Diab et al., 2010; Al-Sabbagh and Girju, 2012; Sadat et al., 2014; Har-rat et al., 2014; Jarrar et al., 2016; Khalifa et al., 2016; Al-Twairish et al., 2018; Alsarsour et al., 2018; Abu Kwaik et al., 2018; El-Haj, 2020; Haff et al., 2022; Nayouf et al., 2023; Jarrar et al., 2023). Several works introducing multi-dialectal datasets and models for region-level dialect identification (Zaidan and Callison-Burch, 2011; Elfardy et al., 2014; Bouamor et al., 2014; Meftouh et al., 2015) and the VarDial workshop series employing transcriptions of speech broadcasts (Malmasi et al., 2016) also followed. Other work developed relatively small-sized commissioned data (Bouamor et al., 2018; Salameh et al., 2018; Obeid et al., 2019).

Subsequently, larger datasets that cover between 10 to 21 countries were introduced (Mubarak and Darwish, 2014; Abdul-Mageed et al., 2018; Zaghoulani and Charfi, 2018; Abdul-Mageed et al., 2020; Abdelali et al., 2021; Issa et al., 2021; Baimukan et al., 2022; Althobaiti, 2022; Elleuch et al., 2025; Hamad et al., 2025). The majority of these datasets are compiled from social media posts, especially X (formerly Twitter). More recently, benchmarks such as ORCA (Elmadany et al., 2023b) and DOLPHIN (Nagoudi et al., 2023) boast dialectal coverage.

Spoken dialect ID shares with text-based dialect ID a scarcity of labeled data. Important efforts to counter this include the introduction of the multi-genre and multidialectal ADI-5 (Ali et al., 2017) and ADI-17 (Ali et al., 2019; Shon et al., 2020) datasets (covering coarse regional and fine-grain country-level dialects, respectively). Moving from text to speech as a modality, however, introduces additional complexities such as potential channel mismatch between train and test sets due to dif-

ferences in recording conditions, as is in the case with ADI-5 (Ali et al., 2017). Furthermore, dialect ID models may capture non-linguistic information such as gender and channel features (Chowdhury et al., 2020), and may experience major performance degradation in cross-domain and cross-dialect settings (Sullivan et al., 2023; Hamad et al., 2025).

2.3 Automatic Speech Recognition

Arabic ASR systems often struggle with dialectal speech, primarily due to lack of (or limited) dialectal data (Waheed et al., 2023). Mozilla Common Voice (Ardila et al., 2020) and MASC (MSA and Dialectal Speech) (Al-Fetyani et al., 2022) were introduced to alleviate this issue. However, both of these corpora label the data under a single label (Arabic) instead of different dialect names. Some of the audio and text samples in these datasets are also misaligned (Lau et al., 2025). The Casablanca project (Talafha et al., 2024) compiled high quality multidialectal speech for eight countries, providing a significant boost towards research in multidialectal ASR. Djanibekov et al. (2025) have also recently presented strong results for dialectal Arabic ASR as well as training strategies that work best based on data availability for each dialect.

2.4 Diacritic Restoration

Several text-based approaches (Alasmary et al., 2024; Elgamal et al., 2024; Fadel et al., 2019; Har-rat et al., 2013) and resources (Toyin et al., 2025; Zerrouki and Balla, 2017) have been proposed for Arabic diacritic / vowel restoration. Aldarmaki and Ghannam (2023) found the speech based approach to outperform text only approaches. More recently, speech based or multi-modal approaches have also been proposed -albeit at a slow rate, mainly due to lack of parallel speech-text resources (Shatnawi et al., 2024). Elmadany et al. (2023a) report strong diacritization models as part of the Octopus toolkit, based on simple finetuning of AraT5 (Elmadany et al., 2022). Shatnawi et al. (2024) also propose an ASR-based diacritic restoration framework, where a pretrained ASR model generates vowelized transcripts refined by a secondary diacritization model. While their approach achieved high accuracy for CA, it fails to generalize to dialectal Arabic due to dataset limitations.

3 NADI 2025

NADI 2025 is the sixth edition of NADI shared task series. Since we extend the scope of the shared task to address broader challenges in multidialectal Arabic speech processing, we refer to NADI 2025 as “the first multidialectal Arabic speech processing shared task”. This edition comprises three complementary subtasks: spoken Arabic dialect identification, multidialectal Arabic ASR, and diacritization restoration. Collectively, these subtasks target critical components of the Arabic speech technology pipeline, each addressing long-standing challenges arising from the language’s rich dialectal variation, frequent code-switching, and the absence of diacritics in most written Arabic. By curating diverse, high quality datasets and establishing standardized evaluation protocols, NADI 2025 aims to catalyze the development of robust, generalizable systems that advance state of the art in Arabic speech and language processing.

3.1 Subtask 1 - Spoken Dialect Identification

Task Description. This subtask is an 8-way classification task to identify which of country-level dialect is being spoken in an utterance, with our set of countries being *Algeria, Egypt, Jordan, Mauritania, Morocco, Palestine, United Arab Emirates (UAE), and Yemen*.

Data. In this subtask, we follow similar procedure in selecting utterances to the data collection procedure of Casablanca (Talafha et al., 2024). For each dialect, different series were identified and the dialect spoken was verified by fluent speakers. For the Adaptation set, we utilize the same series as in Casablanca, but ensure there is no overlap with the series used for the Test set. By doing so, we aim to minimize the influence of potentially overlapping speakers, and to try to disentangle the dialect ID task from simple domain classification.

Evaluation Metric. We use both accuracy as well as the Language Recognition Evaluation 2022 average Cost metric (C_{avg}) (Lee et al., 2022). Because Cost is based on the probability of missed detections as well as false alarms for a given system it provides a complementary way to characterize model performance. At a high level, for two models that have similar accuracy but different Cost, the lower Cost model will providing a larger positive margin between the probability of the correct classes in comparison to incorrect classes, while

the higher Cost model would have a smaller margin between correct and incorrect class probabilities.

3.2 Subtask 2 - Multidialectal Arabic ASR

Task Description. The ASR subtask2 in NADI-2025 focuses on building speech recognition systems that can handle spoken Arabic across a range of regional dialects: *Algerian, Egyptian, Jordanian, Mauritanian, Moroccan, Palestinian, Emirati, and Yemeni*. The task includes both monolingual and code-switched speech, which captures the variation speakers naturally use in different settings.

Data. The dataset used in this subtask is a subset of the Casablanca corpus (Talafha et al., 2024). In this subtask, we select balanced samples from each dialect. The training set is intended primarily for adaptation rather than full model training, encouraging participants to leverage transfer learning, domain adaptation, and other data-efficient strategies. We provide a total of 47,027 utterances, evenly distributed across the eight dialects for the training, validation, and test sets (1,600 utterances per dialect per split). The only exceptions are Algeria, Palestine, and Yemen, which have 727, 900, and 1,180 utterances, respectively, in the test set. These lower counts are due to the limited availability of samples for these dialects in the original Casablanca dataset.

Evaluation Metric. System performance is evaluated using the word error rate (WER) as the primary metric, reported both overall and per dialect. We also report character error rate (CER)⁴ for additional insight into system performance, particularly for short utterances and morphologically rich forms. During evaluation, in line with Talafha et al., 2024, we apply a consistent text normalization pipeline to both system outputs and reference transcripts. Specifically, we: (a) retain only the % symbol, removing other special characters, (b) eliminate diacritics, (c) normalize Hamzas and Maddas to bare alif (ا), (d) convert Eastern Arabic numerals to Western Arabic numerals (e.g., ٢٩ becomes 29), and (e) preserve all Latin characters, as Casablanca contains code-switching segments in other languages.

Subtask 3 - Diacritic Restoration for Spoken Arabic Varieties.

This subtask aims to advance

⁴In the case of a tie, we use the average CER as the tiebreaker.

Dataset	Type	Diacritized	Train	Dev	Test (Ours)
MDASPC	Multi-dialectal	True	60,677	—	5,164
TunSwitch	Dialectal, CS	True	5,212	165	110 (110)
ArzEn	Dialectal, CS	False	3,344	1,402	1,470 (104)
Mixat	Dialectal, CS	False	3,721	—	1,583 (100)
CIArTTS	CA	True	9,500	—	204
ArVoice	MSA	True	2,507	258	(11)
MGB2	MSA	False	—	—	5,365 (40)

Table 1: Number of sentences in datasets provided for the diacritic restoration sub-task. **Ours.** refers to the held-out test set for this shared task which we manually diacritize. **CA.** refers to Classical Arabic. **CS.** refers to code-switching.

research on automatic diacritic restoration for spoken Arabic varieties. As the vast majority of existing vowelization or diacritic restoration efforts focus on CA or MSA, we aim to raise attention to more challenging spoken varieties, such as dialects and code-switching, with a focus on generalization across different varieties. The objective of this sub-task is to restore the diacritics of a given text. The text can be in a variety of forms, including MSA and Arabic dialects and may even include code-switched instances. In addition to text, all inputs have an associated speech utterance to encourage multi-modal approaches.

Data. This subtask encourages the development of multi-modal (speech + text) diacritic restoration models that generalize across Arabic variants. To enable the development of such models, we identified several high-quality data sets (Almeman et al., 2013; Abdallah et al., 2023; Al Ali and Aldarmaki, 2024; Hamed et al., 2020; Kulkarni et al., 2023; Toyin et al., 2025) of Arabic variants (CA, MSA, dialectal, CS) that include parallel speech and text. Table 1 shows a summary of the data sets provided to the participants for this subtask. The MDASPC dataset contains multi-dialectal speech with diacritized transcriptions and we include it for training. For the *TunSwitch* (Abdallah et al., 2023) training data, we used GPT-4o with a chain-of-thought prompt to initially diacritize the transcriptions. The diacritized output of GPT-4o was subsequently manually corrected with the corresponding audio as a reference by a native Arabic speaker. For code-switching, we provide undiacritized resources for training; *ArzEn* (Hamed et al., 2020), *Mixat* (Al Ali and Aldarmaki, 2024) and *MGB2* (Ali et al., 2016); for each dataset, we provide diacritized test sets by manually annotating random subsets of their test set.

Evaluation Metric. Similar to subtask 2, we use WER and CER as performance metrics for this subtask, which are chosen to enable the evaluation of diacritic restoration performance even for models that may change the underlying text, such as ASR-based or sequence-to-sequence models.

4 Shared Task Teams & Results

4.1 Participating Teams

A total of 44 teams registered for the NADI 2025. At the testing phase, a total of 100 valid entries were submitted by *eight* unique teams. The breakdown across the subtasks as follow: 34 submissions for subtask 1 by *five* teams, 47 submissions for subtask 2 by *six* teams and 19 submissions by *two* teams for subtask 3. Table 2 list NADI 2025 participated teams which completed the testing phase.

4.2 Baselines

We developed baseline (BL) models for each subtask to serve as reference points for evaluating the teams’ systems. These models were not shared with participants during the competition.

Subtask 1. We finetune SpeechBrain’s VoxLingua107 (Valk and Alu    , 2021) ECAPA-TDNN (Desplanques et al., 2020) system⁵ on the adaptation split of the dataset. We replace the classification layers of the pretrained system with new randomized layers corresponding to the smaller number of output classes (8); and train these new layers with the rest of the model frozen for 5K steps, and then unfreeze the model and train for an additional 25K steps. We use AdamW with base learning rate of $1e - 4$, and apply a linear ramp up from $1/3$ base LR over 3K steps followed by constant LR until unfreezing, and then repeat the linear ramp up and plateau. Finally, Starting at 20K steps we applying an exponential decay.

Subtask 2. A zero-shot baseline is built on the pre-trained Whisper-Large-v3 model (Radford et al., 2022). Dialect-wise inference is performed on the official NADI 2025 subtask 2 ASR release available on Hugging Face⁶, which provides validation splits for eight country-level dialects; official evaluation is conducted on a private Codabench test set. During inference, audio inputs are transcribed

⁵<https://huggingface.co/speechbrain/spkrec-ecapa-voxceleb>

⁶https://huggingface.co/datasets/UBC-NLP/NADI2025_subtask2_ASR

Team Name	Affiliation	Subtask
Abjad AI (Ghannam et al., 2025)	Abjad AI, Jordan & Saudi Arabia	1,3
BYZÖ (Abdullah et al., 2025)	Saarland University, Germany	2
Elyadata (Elleuch et al., 2025)	Elyadata, Tunisia	1,2
Hamsa	Hamsa	2
Lahjati (ALBawwab and Qawasmeh, 2025)	Princess Sumaya University, Jordan	1
MarsadLab (Attia et al., 2025)	Hamad Bin Khalifa University, Qatar	1,2
Munsit (Salhab et al., 2025b)	Lebanese American University, Lebanon	2
Unicorn (Elrefai, 2025)	Ain shams University, Egypt	3

Table 2: List of teams that participated in NADI 2025 shared task. Teams with accepted papers are cited.

using Whisper’s default decoding parameters with language explicitly set to Arabic.

	Accuracy $\uparrow$	C_{avg} $\downarrow$
ELYDATA-LIA (Elleuch et al., 2025)	79.8	17.88
BYZÖ-ADI (Abdullah et al., 2025)	76.4	22.65
MarsadLab (Attia et al., 2025)	61.6	30.68
Abjad AI	61.2	34.77
Baseline	61.1	34.22
Lahjati (ALBawwab and Qawasmeh, 2025)	50.8	48.99

Table 3: Performance of the systems on the test set for Subtask 1. Results are sorted by Accuracy, while the average cost (C_{avg}) score is also reported, with lower values indicating better performance. The best performance is highlighted in bold.

Subtask 3. In this subtask, we provide three baselines: (I) A *text only* baseline based on the publicly available CATT model (Alasmary et al., 2024), which we use without further fine-tuning (II) an *ASR based* baseline where we use the ArTST v3 checkpoint (Djanibekov et al., 2025), which is pre-trained on dialectal and code-switched Arabic, and finetune it for Arabic ASR with diacritics using the provided training data, and (III) a *multi-modal* diacritc restoration model designed as follows:

The raw waveform and corresponding transcriptions are passed in parallel to a speech and text encoder, respectively. The speech encoder is derived from ArTST ASR (Djanibekov et al., 2025), and the text encoder is derived from ArTST TTS model (Toyin et al., 2023). We then align the resulting text and speech embeddings using multi-head attention with 8 heads, followed by a trainable prediction component comprising 2 bi-directional LSTM layers, a 30% dropout layer, and a final linear prediction head to predict the corresponding diacritics. Simple ad-hoc post-processing is applied to add the predicted diacritics to the input text to produce the fully diacritized text output. This approach is inspired by the multi-modal diacritization model described in Shatnawi et al. (2024).

4.3 Results

Tables 3, 4, and 5, present the preformance of the submitted systems on the test set for subtask 1, subtask 2, and subtask 3 respectively.

Subtask1. The ELYDATA-LIA team (Elleuch et al., 2025) achieved the best performance in terms of both accuracy and average cost C_{avg} (79.8 / 17.88), followed closely by BYZÖ-ADI (Abdullah et al., 2025) (76.4 / 22.65). Both top teams addressed the limited size of the Adaptation set in novel ways: ELYDATA-LIA leveraged the much larger ADI-20 dataset (Elleuch et al., 2025), while BYZÖ-ADI employed kNN voice conversion (Baas et al., 2023) to augment the training data with synthetic samples. In third place, MarsadLab (Attia et al., 2025) improved upon the baseline system through additional data augmentation and the introduction of an attention mechanism prior to the classification layer. In fourth place, Abjad AI fine-tuned a Whisper Small encoder with further data augmentation. While the third- and fourth-place systems were close in terms of accuracy (61.6 vs. 61.2), the approach by MarsadLa achieved a notably better C_{avg} , reducing it by approximately 4 points. Finally, we note that one team (Lahjati (ALBawwab and Qawasmeh, 2025)) perform below the baseline. Overall, these results highlight the effectiveness and diversity of data augmentation strategies.

Subtask 2. The Munist team (Salhab et al., 2025b) obtain the lowest overall average WER/CER scores (35.68/12.10) among all participating systems, achieving the best performance across all dialects except Moroccan, where it ranked second in both WER and CER, and Mauritanian, where it ranked first in CER and second in WER. The ELYADATA-LIA team (Elleuch et al., 2025) ranked second with scores of 38.52/14.52. They achieved the best performance on the

	Average	JOR	EGY	MOR	ALG	YEM	MAU	UAE	PAL
Munsit (Salhab et al., 2025b)	35.68/12.20	20.68/5.64	20.88/7.33	41.71/14.04	53.62/18.44	44.62/14.30	59.03/23.28	22.66/6.55	22.27/8.05
ELYADATA-LIA (Elleuch et al., 2025)	38.53/14.52	28.03/9.36	26.83/11.43	38.26/13.66	53.73/20.43	46.63/16.66	58.10/24.53	29.35/9.91	27.36/10.20
BYZÖ-Whisper (Abdullah et al., 2025)	39.78/14.75	28.84/9.47	29.50/11.91	43.06/15.52	55.04/20.59	46.41/16.05	59.36/24.84	28.38/9.04	27.65/10.59
Hamsa	42.04/16.18	32.24/9.90	24.72/10.21	48.21/18.11	60.32/23.33	51.76/20.41	66.23/29.11	28.00/8.98	24.87/9.41
BYZÖ-CTC (Abdullah et al., 2025)	44.14/15.58	31.74/9.94	37.23/12.57	43.31/15.07	56.12/21.38	46.14/15.68	63.32/26.70	38.65/11.14	36.62/12.18
Baseline	93.89/72.79	46.09/19.28	100.06/81.37	100.38/80.42	101.03/79.58	101.09/80.58	100.59/82.89	101.15/80.27	100.76/77.92
MarsadLab (Attia et al., 2025)	104.89/84.69	44.97/19.19	113.97/97.65	104.07/87.58	116.59/94.26	113.54/94.56	111.59/92.84	116.79/97.00	117.60/94.42

Table 4: Performance of the systems on the test set for Subtask 2. Results are sorted by the overall average WER/CER score across all dialects, with lower values indicating better performance. The best performance is highlighted in bold.

	WER ↓	CER ↓
Abjad AI (Ghannam et al., 2025)	55	13
Unicorn (Elrefai, 2025)	64	15
Baseline-I (ASR based)	88	45
Baseline-II (text-only)	65	16
Baseline-III (multi-modal)	66	16

Table 5: Performance of the systems on the test set for Subtask 3. Results are sorted by the overall average WER/CER score across all dialects, with lower values indicating better performance. The best performance is highlighted in bold.

Moroccan dialect (WER/CER of 38.26/13.66) and obtain the lowest CER for the Mauritanian dialect. Their performance on the Algerian dialect was only marginally lower than that of the first-ranked team, suggesting that their system demonstrates strong capabilities for North African dialects in general. The BYZÖ-Whisper team (Abdullah et al., 2025) ranked third, with average WER/CER scores of 39.78/14.75. The Hamsa team follow in fourth place, scoring 42.04/16.18, while the BYZÖ-CTC team (Abdullah et al., 2025) ranked fifth with 44.14/15.58. Only one team, MarsadLab (Attia et al., 2025), perform below the baseline, with notably higher average WER/CER scores of 104.89/84.69. The winning team Munist (Salhab et al., 2025b) surpassed the baseline by **58.21** WER points (93.89 $\rightarrow$ 35.68; $\approx$ 62% reduction). Furthermore, the variation in *WER* scores among the teams that surpassed the baseline is relatively low ($\sigma \approx 3.25$), corresponding to about 8.1% of the mean WER for these systems.

Subtask 3. The Abjad AI (Ghannam et al., 2025) perform the best with the lowest WER of 55% and CER of 13%. The Unicorn team (Elrefai, 2025) follow closely behind with WER of 64% and CER of 15%. Both teams improve over the provided baselines, the best of which achieve WER and CER of 65% and 16%, respectively.

5 Overview of Submitted Systems

In this section, we present an overview of the submitted systems for each subtask and summarize the methodological approaches adopted by the participating teams.

5.1 Subtask 1

ELYADATA-LIA (Elleuch et al., 2025). Using Whisper Large-v3 encoder as their base model, they adopt a two stage finetuning procedure to first finetune on the forthcoming ADI-20 dataset (Elleuch et al., 2025), and then use the NADI ADI Adaptation set for a second finetuning. Features of this approach include freezing the first 16 layers of the encoder and using plenty of data augmentation methods including speed perturbation, added noise, and frequency and chunk dropping.

BYZÖ-ADI (Abdullah et al., 2025) The authors choose a straightforward finetuning approach using w2v-BERT-2.0 (Barrault et al., 2023) model finetuned on the NADI ADI split (69% accuracy). However, in order to improve the robustness of the model, they add a data augmentation approach by using a voice conversion model (Baas et al., 2023) to re-synthesizing the training utterances using voice samples from the 4 Arabic speakers from the LibriVox project, and training on the mixed natural and synthetic audio, leading to their final model.

MarsadLab (Attia et al., 2025) Adopts a starting point similar to the baseline with a VoxLingua107 ECAPA-TDNN system that was finetuned on the ADI task. They introduce a number of features in the process including feature reweighting of the hidden representation just prior to the classification layer through the use of a lightweight attention mechanism, discriminative learning rate of the classification head, progressive unfreezing, as well as data augmentation using SpecAugment and injected noise.

Abjad AI Like the ELYDATA-LIA approach, this team used Whisper model, Whisper Small, and finetuned the encoder for dialect ID. They use only the NADI Adaptation set for finetuning, using SpecAugment (time and frequency masking) for data augmentation, and unfreezing the model partway through training.

Lahjati (ALBawwab and Qawasmeh, 2025) Using both the VoxLingua107 ECAPA-TDNN system as well as WavLM encoder, this fusion approach concatenates the outputs from the two models (WavLM pooled to match the ECAPA 256 dimension), and passes this combined representation through a layer normalization layer and then a two layers feedforward network to perform classification. Similar to other approaches the underlying models start frozen, with unfreezing at 8000 steps, followed by a ramp up, plateau, and then cosine annealing learning rate schedule.

5.2 Subtask 2

Munsit (Salhab et al., 2025b) This system follows a two-stage training pipeline combining large-scale weakly supervised pretraining and continual supervised fine-tuning, inspired by Salhab et al., 2025a. In the first stage, a Conformer-large model (Gulati et al., 2020) (121M parameters) was pretrained on 15K hours of weakly labeled Arabic speech, covering MSA and various dialects, with automatic labeling and no manual verification. In the second stage, the model was fine-tuned using a high-quality dataset composed of 3,000 hours of rigorously filtered weakly labeled data, excluding news content, and the official Casablanca Challenge training set, expanded via data augmentation. Training used the CTC (Graves et al., 2006) objective with a SentencePiece (Kudo and Richardson, 2018) vocabulary of 128 tokens, AdamW optimizer, Noam learning rate schedule, and dropout of 0.1, in a distributed setup across 8 NVIDIA A100 GPUs with bfloat16 precision. This approach enabled robust performance across all dialects, achieving the lowest average WER and CER in the shared task.

ELYADATA & LIA (Elleuch et al., 2025) For the ASR subtask, this team fine-tuned the SeamlessM4T-v2 (Barrault et al., 2023) Large Egyptian model separately for each of the eight dialects in the Casablanca dataset, producing eight distinct models. Training was performed for 6 epochs on NVIDIA A100 GPUs with a label-smoothed NLL loss (smoothing 0.2), AdamW opti-

mizer, and a learning rate schedule with 100 warm-up steps ramping from 1e-9 to 5e-5. A batch size of 2 was used for all runs. This per-dialect fine-tuning approach yielded second overall in the shared task.

BYZÖ (Abdullah et al., 2025) The team submitted two independent systems. The first, BYZÖ-Whisper, fine-tuned the Whisper-Large-v3 model (Radford et al., 2023) (1.54B parameters) for Arabic dialect ASR using only the NADI shared task data, without external datasets or data augmentation. Text labels were preprocessed by removing bracketed content and normalizing spacing. Training followed a two-stage process: (1) domain adaptation on combined data from all dialects for 9000 steps (learning rate 1e-5, batch size 32), and (2) dialect-specific adaptation for 2000 steps per dialect using CER as the metric. The second, BYZÖ-CTC, fine-tuned the w2v-BERT-2.0 model (Barrault et al., 2023) (580M parameters) using a mix of public Arabic ASR datasets, then further fine-tuned per dialect on the shared task data (learning rate 1e-5, batch size 16). A multi-dialectal 3-gram Kneser-Ney smoothed language model, trained on collected dialect-specific text data, was integrated to reduce WER. This encoder-only CTC-based system was noted for efficiency and competitive zero-shot performance compared to Whisper large.

MarsadLab (Attia et al., 2025) For the ASR subtask, this team adopted Whisper-Large model (Radford et al., 2023) in a zero-shot setting, without any fine-tuning, preprocessing, or post-processing. Leveraging Whisper’s multilingual capabilities, the system directly transcribed Arabic speech from multiple dialects in the test set. While the ECAPA-TDNN architecture was central to their ADI submission, it was not applied to ASR.

Hamsa Submissions were received from the Hamsa team; however, a system description was not made available.

5.3 Subtask 3

Unicorn (Elrefai, 2025) This team addressed the diacritic restoration task by fine-tuning the GEMM3N⁷ multimodal model on both audio and text inputs. They formed diacritic restoration as a structured generation task where the model receives an undiacritized sentence and its corresponding audio and generates a fully diacritized ver-

⁷<https://unsloth.ai/>

sion. They fine-tuned with LoRA adaptation to efficiently adapt the model with the provided data for the sub-task only. They applied *nlpaug* for speech augmentation to simulate more diverse audio inputs. They perform inference by prompting GEMM3N with the raw audio and the corresponding undiacritized text.

Abjad AI (Ghannam et al., 2025) This team presented CATT-Whisper, which is a multimodal approach that combines both textual and speech information. Their model represents the text modality using an encoder extracted from their pre-trained model named CATT (Alasmmary et al., 2024). The speech component is handled by the encoder module of the OpenAI Whisper base model (Radford et al., 2022). Their approach uses two integration strategies. The former consists of fusing the speech tokens with the input at an early stage, where the 1500 frames of the audio segment are averaged on the basis of 10 consecutive frames, resulting in 150 speech tokens only. To ensure embedding compatibility, these averaged tokens are processed through a linear projection layer prior to merging them with the text tokens. Contextual encoding is guaranteed by the CATT encoder module. The latter strategy relies on cross-attention, where text and speech embeddings are fused. Then, finally, the cross-attention output is fed to the CATT classification head for token-level diacritic prediction. They randomly deactivate the speech input during training for robustness, which allows the model to perform well with or without speech.

6 Conclusion

The *sixth* NADI shared task extends the scope of the series beyond text-based processing to encompass speech and diacritization, introducing three new subtasks: spoken dialect identification, Arabic ASR, and diacritic restoration. By releasing high-quality resources and providing clear evaluation protocols, our goal is to foster progress in inclusive Arabic speech processing. This edition, we received 44 registrations, with *eight* teams submitting system outputs and *seven* system description papers accepted. Results across the three subtasks highlight substantial headroom for improvement: even strong pretrained models continue to face challenges with multidialectal variability, code-switching, and diacritic restoration. We hope that this edition not only advances the state of the art on each individual subtask but also inspires fu-

ture research toward unified, dialect-aware speech technologies for Arabic.

Limitations & Ethical Considerations

Despite the contributions of this year’s shared task, several limitations remain across the three subtasks:

Coverage of dialects: Not all Arabic dialects are represented in the test sets, which limits the generalizability of results across the full dialect continuum.

Country-level labeling: We acknowledge that the use of country-level labels may be problematic. The continuum of Arabic dialects is complex, and using country affiliation as a stand-in for well-defined linguistic boundaries is not without limitations. This choice was made to ensure a reasonable degree of diversity in dialect coverage, while avoiding assumptions about the generalizability of models trained on a subset of dialects to unseen but related varieties.

Code-switching: The datasets capture only a limited subset of code-switching phenomena, whereas real-world Arabic speech often involves more diverse language mixing.

Real-world conditions: Background noise, spontaneous disfluencies, and accented speech are underrepresented in the datasets, limiting ecological validity.

Evaluation metrics: Metrics such as WER and CER may be misleading in the ASR task, since a dialectal utterance can often have multiple valid references. As the data provides only one reference per utterance, evaluation scores may underestimate system performance by penalizing alternative but correct transcriptions.

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⁸<https://alliancecan.ca>

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